21 Mechanical testing

21.1 Hardness testing

Hardness tests are a quick and simple method to characterise a metal's resistance to deformation and can in some cases be related to an equivalent yield strength. A number of different hardness tests exist, which differ in the type of indenter used, the load applied, and the means used to quantify the extent of permanent deformation. Further information concerning the hardness of metals and hardness testing may be found in Refs [1–3]. Hardness values for some common metals and alloys are given in Chapter 22.

21.1.1 Brinell hardness

Relevant standards: ASTM E10-01, BS EN 10003-1:1996
An indenter comprising a hardened steel ball of diameter D mounted in a suitable holder is pushed into the material under test by a force F. The diameter of the indentation left in the surface of the material after removal of the load is measured in two directions at right angles. The area of the curved surface of the indentation is calculated from the mean diameter, d, the indentation being considered as a segment of a sphere of diameter D. The Brinell hardness is the quotient obtained by dividing the load, symbol f, expressed in kilograms-force, by the surface area of the indentation expressed in square millimetres.

Symbols:

F = force (N)
f = load symbol (kgf)
D = diameter of ball in millimetres (mm)
d = mean diameter of indentation in millimetres (mm)

$$\text{HBW} = \text{Brinell Hardness} = \frac{0.102 \times 2F}{\pi D[D - \sqrt{D^2 - d^2}]} = \frac{2f}{\pi D[D - \sqrt{D^2 - d^2}]}$$

$$h = \text{depth of indentation in millimetres} = \frac{f}{\pi D(\text{HBW})} \text{ (mm)}$$

The symbol HBW is supplemented by numbers indicating the diameter of the ball used and the load applied. Thus 226 HBW 10/3 000 indicates that a Brinell hardness of 226 was obtained by using a 10 mm diameter ball with a load symbol of 3 000 kgf. If the time of duration of load differs from the standard 10–15 sec, a further number is added to show the duration of the load in seconds.

RELATION OF LOAD TO BALL DIAMETER

The choice of force and ball diameter to be used in a Brinell test is determined by two factors:

1. the value of the ratio f/D^2, and
2. the size of the indentation which provides optimum accuracy.

The same value of f/D^2 will, in principle, give the same hardness value for different loads and the load used in practice depends on the nature and the hardness of the material under test. Five standard values of f/D^2, i.e. 30, 10, 5, 2.5 and 1, have been adopted. Recommended combinations of load and diameter for different materials are given in BS EN 10003-1:1996.

21.1.2 Rockwell hardness

Relevant standards: ASTM E18-00, BS EN ISO 6508-1:1999

An indenter of the standard type comprising a diamond cone or hardened steel ball mounted rigidly in a suitable holder, is forced into the test piece under a preliminary load F_0. When equilibrium has been reached, an indicating device which follows the movement of the indenter and so responds to changes in depth of penetration of the indenter is set to a datum position.

While the preliminary load is still applied, it is augmented by an additional load with resulting increase in penetration of the indenter. When equilibrium has again been reached, the additional load is removed but the preliminary load is maintained.

Removal of the additional load allows a partial recovery, so reducing the depth of penetration. The permanent increase in depth of penetration, e, resulting from application and removal of the additional load is used to deduce the Rockwell hardness number by means of the equation:

$$\text{HR} = E - e$$

where:

HR = Rockwell hardness number
E = a constant depending on the form of indenter:
100 units when a diamond cone indenter is used
130 units when a steel ball indenter is used

A variety of indenters and loads are used to give several scales of hardness, e.g. 60 HRC represents a Rockwell hardness of 60 on scale C. The various scales are defined in Table 21.1.

Table 21.1 ROCKWELL HARDNESS SCALES

Scale symbol		*Total test force* kN	kgf	*Dial figures*	*Typical applications of scales*
B	$\frac{1}{16}$ in (1.588 mm) ball	0.98	100	Red	Copper alloys, soft steels, aluminium alloys, malleable iron, etc.
C	Diamond	1.47	150	Black	Steel, hard cast irons, pearlitic malleable iron, titanium, deep case hardened steel and other materials harder than HRB100
A	Diamond	0.59	60	Black	Cemented carbides, thin steel, and shallow case-hardened steel
D	Diamond	0.98	100	Black	Thin steel and medium case hardened steel, and pearlitic malleable iron
E	$\frac{1}{8}$ in (3.175 mm) ball	0.98	100	Red	Cast iron, aluminium and magnesium alloys, bearing metals
F	$\frac{1}{16}$ in (1.588 mm) ball	0.59	60	Red	Annealed copper alloys, thin soft sheet metals
G	$\frac{1}{16}$ in (1.588 mm) ball	1.47	150	Red	Malleable irons, copper–nickel–zinc and cupro–nickel alloys. Upper limit HRG 92 to avoid possible flattening of ball
H	$\frac{1}{8}$ in (3.175 mm) ball	0.59	60	Red	Aluminium, zinc, lead
K	$\frac{1}{8}$ in (3.175 mm) ball	1.47	150	Red	Bearing metals and other very soft or thin materials. Use smallest ball and heaviest load that does not give anvil effect
L	$\frac{1}{4}$ in (6.350 mm) ball	0.59	60	Red	
M	$\frac{1}{4}$ in (6.350 mm) ball	0.98	100	Red	
P	$\frac{1}{4}$ in (6.350 mm) ball	1.47	150	Red	
R	$\frac{1}{2}$ in (12.70 mm) ball	0.59	60	Red	
S	$\frac{1}{2}$ in (12.70 mm) ball	0.98	100	Red	
V	$\frac{1}{2}$ in (12.70 mm) ball	1.47	150	Red	

Notes: (1) The preliminary load is 10 kgf for all scales.
(2) The diamond indenter is conical with an included angle of 120 ± 0.1° with a tip rounded to a radius of 0.20 mm.
(3) ASTM E18-00 standard gives values in kgf. Newton values are conversions.

21.1.3 Rockwell superficial hardness

Relevant standards: ASTM E18-00, BS EN ISO 6508-1:1999
This test is similar in principle to the Rockwell test, but in order to keep the depth of impression small the minor load is restricted to 3 kg. It is used for thin specimens and for nitrided steels, etc. The following scales are used.

Table 21.2 ROCKWELL SUPERFICIAL HARDNESS SCALES

		Preliminary Test Force		*Total Test Force*	
Scale	*Penetrator*	kgf	N	kgf	kN
15-N	Diamond cone	3	29.4	15	0.14
30-N	Diamond cone	3	29.4	30	0.29
45-N	Diamond cone	3	29.4	45	0.44
15-T	$\frac{1}{16}$ in (1.588 mm) steel ball	3	29.4	15	0.14
30-T	$\frac{1}{16}$ in (1.588 mm) steel ball	3	29.4	30	0.29
45-T	$\frac{1}{16}$ in (1.588 mm) steel ball	3	29.4	45	0.44

21.1.4 Vickers hardness test

Relevant standards: BS EN ISO 6507-1:1998, ASTM E92-82 (1997)
A diamond indenter, in the form of a right pyramid with a square base and an angle of 136° between opposite faces, is forced into the material under a load F. The two diagonals, d_1 and d_2 of the indentation left in the surface of the material after removal of the load are measured and their arithmetic mean d calculated. The area of the sloping surface of the indentation is calculated, the indentation being considered as a right pyramid with a square base of diagonal d and vertex angle of 136°.

The Vickers hardness is the quotient obtained by dividing the load F, expressed in kilograms-force, by the sloping area of the indentation expressed in square millimetres.

Symbols:

F = load in kilograms force (kgf) where 1 kgf = 9.806 65 N
d = arithmetic mean of the two diagonals d_1 and d_2 in millimetres (mm)
HV = Vickers hardness

$$= 2F \sin(136°/2)d^2$$
$$= 1.854F/d^2$$

The loads employed vary from 1 to 100 kgf and are maintained from 10 to 15 seconds. Hardness numbers are expressed as 440 HV30 for example where a hardness of 440 is measured using a 30 kgf load.

21.1.5 Micro-indentation hardness testing

Relevant standards: ASTM E 384-99, BS EN ISO 6507-1:1997
Micro-indentation hardness testing is performed according to principles similar to the Vickers hardness test, but with considerably lower loads that produce microscopic indentations (typically 10–200 μm). Micro-indentation hardness testing, also referred to as 'microhardness' testing, is commonly used to measure hardness in very local areas, e.g. different microstructural phases or surface layers. As in the Vickers test, a diamond indenter with a specific geometry is impressed into the specimen surface with a known, calibrated force for a duration of 10–15 sec. Forces range from 1–1 000 gf. The indenter is removed from the surface and the size of the impression is measured using a microscope. Micro-indentation hardness testers are sold commercially.

Two indenter shapes are employed. The Vickers indenter is a square-based pyramid with face angles of 136°. The lengths of the two diagonals of the resulting square indent are averaged and the hardness, HV, is calculated according to:

$$HV = 1\,854.4\,P/d^2$$

where P is the applied force in gf and d is the mean diagonal length.

The Knoop indenter is a rhombic-based pyramid with included longitudinal edge angles of 172.5° and 130°; the long diagonal is approximately seven times the length of the short diagonal. The long diagonal is measured, and the hardness, HK, is calculated according to:

$$HK = 14\,229\,P/d^2$$

The narrow width of the Knoop indent makes it better suited for measuring steep hardness gradients since the indents can be more closely spaced.

Vickers hardness values are reported in the form: *value* HV_{force}, for example, 650 HV_{100} corresponds to a Vickers hardness of 600 measured with a 100 gf force. The same form is used for Knoop hardness with HK instead of HV. In the European standard, force is reported in units of kgf.

When selecting the force to be used for testing, it is important to recognise that the measured hardness is not necessarily independent of test force, especially for forces less than 100 gf. Larger indents produce better accuracy due to reduced error in length measurement. The size of the indent, however, must be considerably smaller than the area of interest. In addition, the indenter must be perpendicular to the test piece, and the surface being measured must be free of deformation and damage resulting from preparation (e.g. grinding and polishing). Ref. [4] provides further information on microindentation hardness testing.

Recently, the techniques of microhardness indentation testing have been extended to smaller scales using lower loads and more precise instrumentation. This class of testing is referred to as nanohardness testing, nanoindentation testing, or instrumented indentation testing. These test techniques commonly employ a three-sided pyramidal indenter, rather than the four-sided Vickers indenter, and include the ability to measure and record the force on the indenter along with its vertical displacement. The latter feature of nanoindentation testing allows it to used to characterise a number of properties of surfaces and thin films, including elastic modulus, yield stress and strain-hardening behaviour, fracture toughness of brittle materials, and creep behaviour. Ref. [5] provides further information and references related to nanoindentation testing.

21.1.6 Hardness conversion tables

Table 21.3 only applies to steel of uniform chemical composition and uniform heat treatment, and is not recommended for non-ferrous metals or for case-hardened steels.

Table 21.3 APPROXIMATE CONVERSION OF HARDNESS VALUES

Non-austenitic steels

		Rockwell hardness No.			
Vickers hardness No. HV	*Brinell* 3 000 kg *load* 10 mm *ball* HBW	*C Scale* 150 kg *load diamond cone* HRC	*A scale* 60 kg *load diamond cone* HRA	*Superficial* 30–*N* 10 kg *load diamond cone*	*Scleroscope hardness No.*
100	95	—	43	—	—
120	115	—	46	—	—
140	135	—	50	—	21
160	155	—	53	—	24
180	175	—	56	—	27
200	195	—	58	—	30
220	215	—	60	—	31
240	235	20.3	60.7	41.7	34
260	255	24.0	62.4	45.0	37
280	275	27.1	63.8	47.8	40
300	295	29.8	65.2	50.2	42
320	311	32.2	66.4	52.3	45
340	328	34.4	67.6	54.4	47

(*continued*)

Table 21.3 APPROXIMATE CONVERSION OF HARDNESS VALUES—*continued*

		Rockwell hardness No.			
Vickers hardness No. HV	*Brinell* 3 000 kg *load* 10 mm *ball* HBW	*C Scale* 150 kg *load diamond cone* HRC	*A scale* 60 kg *load diamond cone* HRA	*Superficial* 30–*N* 10 kg *load diamond cone*	*Scleroscope hardness No.*
360	345	36.6	68.7	56.4	50
380	360	38.8	69.8	58.4	52
400	379	40.8	70.8	60.2	55
420	397	42.7	71.8	61.4	57
440	415	44.5	72.8	63.5	59
460	433	46.1	73.6	64.9	62
480	452	47.7	74.5	66.4	64
500	471	49.1	75.3	67.7	66
520	487	50.5	76.1	69.0	67
540	507	51.7	76.7	70.0	69
560	525	53.0	77.4	71.2	71
580	545	54.1	78.0	72.1	72
600	564	55.2	78.6	73.2	74
620	582	56.3	79.2	74.2	75
640	601	57.3	79.8	75.1	77
660	620	58.3	80.3	75.9	79
680	638	59.2	80.8	76.8	80
700	656	60.1	81.3	77.6	81
720	670	61.0	81.8	78.4	83
740	684	61.8	82.2	79.1	84
760	698	62.5	82.6	79.7	86
780	710	63.3	83.0	80.4	87
800	722	64.0	83.4	81.1	88
820	733	64.7	83.8	81.7	90
840	745	65.3	84.1	82.2	91
860	—	65.9	84.4	82.7	92
880	—	66.4	84.7	83.1	93
900	—	67.0	85.0	83.6	95
920	—	67.5	85.3	84.0	96
940	—	68.0	85.6	84.4	97

* Brinell values greater than 480 are determined with a carbide ball.

Table 21.4 APPROXIMATE CONVERSION OF HARDNESS VALUES

Aluminium and its alloys

HV	*Brinell**	*Rockwell superficial* HR			HV	*Brinell**	*Rockwell superficial* HR		
10 kg	HBW	15*T*	30*T*	45*T*	10 kg	HBW	15*T*	30*T*	45*T*
20	19.4	2.3	—	—	80	74.6	75.3	44.1	—
24	23.1	18.1	—	—	84	78.3	76.5	46.9	—
28	26.8	29.5	—	—	88	82.0	77.6	49.4	—
32	30.5	38.2	—	—	92	85.7	78.6	51.8	—
36	34.2	45.0	—	—	96	89.4	79.5	54.0	—
40	37.8	50.3	—	—	100	93.0	80.3	55.9	—
44	41.5	54.8	—	—	104	96.7	81.1	57.7	—
48	45.2	58.6	5.2	—	108	100.4	81.8	59.4	—
52	48.9	61.8	12.6	—	112	104.1	82.5	60.9	—
56	52.6	64.5	19.1	—	116	107.8	83.1	62.4	—
60	56.2	66.9	24.5	—	120	111.4	83.7	63.8	39.0
64	59.9	69.0	29.4	—	124	115.1	84.3	65.0	41.2
68	63.6	70.8	33.7	—	128	118.8	84.8	66.2	43.3
72	67.3	72.5	37.6	—	132	122.5	85.3	67.4	45.2
76	71.0	74.0	41.0	—	136	126.2	85.7	68.5	47.1

(*continued*)

Table 21.4 APPROXIMATE CONVERSION OF HARDNESS VALUES—*continued*

HV	*Brinell**	*Rockwell superficial* HR			HV	*Brinell**	*Rockwell superficial* HR		
10 kg	HBW	15*T*	30*T*	45*T*	10 kg	HBW	15*T*	30*T*	45*T*
140	129.8	86.2	69.4	48.8	172	159.3	88.3	75.8	59.6
144	133.5	86.6	70.4	50.4	176	163.0	89.2	76.4	60.7
148	137.2	86.9	71.3	51.9	180	166.6	89.4	77.0	61.7
152	140.9	87.3	72.1	53.3	184	170.3	89.7	77.6	62.7
156	144.6	87.6	72.9	54.7	188	174.0	89.9	78.1	63.6
160	148.2	88.0	73.7	56.0	192	177.7	90.1	78.7	64.5
164	151.9	88.3	74.4	57.3	196	181.4	90.3	79.2	65.4
168	155.6	88.6	75.1	58.4	200	185.0	90.6	79.7	66.2

* The ratio $\frac{\text{load}}{(\text{dia. of ball})^2}$ for the Brinell tests was 5 for Brinell hardnesses of 20–60 and 10 for hardnesses above 60.

Table 21.5 APPROXIMATE CONVERSION OF HARDNESS VALUES

Brass

	Rockwell hardness No.		*Rockwell superficial hardness No.*			*Brinell hardness No.*		*Rockwell hardness No.*		*Rockwell superficial hardness No.*			*Brinell hardness No.*
HV	*B scale*, 100 kg *load*, $\frac{1}{16}$ in *ball*	*F scale*, 60 kg *load*, $\frac{1}{16}$ in *ball*	15-*T scale*, 15 kg *load*, $\frac{1}{16}$ in *ball*	30-*T scale*, 30 kg *load*, $\frac{1}{16}$ in *ball*	45-*T scale*, 45 kg *load*, $\frac{1}{16}$ in *ball*	500 kg *load*, 10 mm *ball*	HV	*B scale*, 100 kg *load*, $\frac{1}{16}$ in *ball*	*F scale*, 60 kg *load*, $\frac{1}{16}$ in *ball*	15-*T scale*, 15 kg *load*, $\frac{1}{16}$ in *ball*	30-*T scale*, 30 kg *load*, $\frac{1}{16}$ in *ball*	45-*T scale*, 45 kg *load*, $\frac{1}{16}$ in *ball*	500 kg *load*, 10 mm *ball*
45	—	40.0	—	—	—	42	120	67.0	95.5	—	61.0	41.0	106
47	—	45.0	—	—	—	44	124	69.0	96.5	—	62.5	43.0	110
49	—	49.0	54.5	—	—	46	128	71.0	97.5	—	63.5	45.0	113
52	—	53.5	57.0	—	—	48	132	73.0	98.5	84.5	65.0	46.5	116
56	—	58.8	60.0	15.0	—	52	136	74.5	99.5	85.0	66.0	48.0	120
60	10.0	63.0	62.5	20.5	—	55	140	76.0	100.5	85.5	67.0	50.0	122
64	15.5	66.8	65.0	22.5	—	59	144	77.8	101.5	86.0	68.0	51.5	126
68	21.5	70.0	67.0	30.0	—	62	148	79.0	102.5	—	69.0	53.0	129
72	27.5	73.2	69.0	34.0	—	64	152	80.5	103.0	—	—	54.0	133
76	32.5	76.0	70.5	38.0	4.5	68	156	82.0	104.0	87.0	70.5	55.5	136
80	37.5	78.6	72.0	41.0	10.0	72	160	83.5	—	—	71.5	56.5	139
84	42.0	81.2	73.5	44.0	14.5	76	164	85.0	105.5	—	72.0	58.0	142
88	46.0	83.5	75.0	47.0	19.0	79	168	86.0	106.0	88.0	73.0	59.0	146
92	49.5	85.4	76.5	49.0	23.0	82	172	87.5	106.5	—	73.5	60.0	149
96	53.0	87.2	77.5	51.5	26.5	85	176	88.5	107.0	—	—	61.0	152
100	56.0	89.0	78.5	53.5	29.5	88	180	90.0	107.5	—	75.0	62.0	156
104	58.0	90.5	79.5	55.0	32.0	92	184	91.0	—	—	75.5	63.0	159
108	61.0	92.0	—	57.0	34.5	95	188	92.0	—	89.5	—	64.0	162
112	63.0	93.0	81.0	58.5	37.0	99	192	93.0	—	—	77.0	65.0	166
116	65.0	94.5	82.0	60.0	39.0	103	196	93.5	110.0	90.0	77.5	66.0	169

Note: $\frac{1}{16}$ in ≡ 1.588 mm.

Table 21.6 APPROXIMATE CONVERSION OF HARDNESS VALUES

Hard metals

Vickers diamond hardness, 50 kg *load*	*Rockwell A scale* 60 kg *load: diamond cone*	*Rockwell C scale* 150 kg *load: diamond cone*
1 750	92.4	80.5
1 700	92.0	79.8
1 650	91.7	79.2
1 600	91.3	78.4
1 550	90.9	77.7
1 500	90.5	77.0
1 450	90.1	76.2
1 400	89.7	75.4
1 350	89.3	74.6
1 300	88.9	73.8
1 250	88.5	73.0
1 200	88.1	72.2
1 150	87.6	71.3
1 100	87.0	70.4
1 050	86.4	69.4
1 000	85.7	68.2
950	85.0	66.6
900	84.0	64.6
850	82.8	—

Table 21.7 APPROXIMATE CONVERSION OF HARDNESS VALUES FOR NICKEL AND HIGH NICKEL ALLOYS

Vickers hardness No. HV	*Brinell hardness No.* HBW	*Rockwell hardness No.*		
		A scale	*B scale*	*C scale*
Vickers indenter 1, 5, 10 *and* 30 kgf	10 mm *ball* 3 000 kgf	*Diamond penetrator* 60 kgf	$\frac{1}{16}$ in *ball* 100 kgf	*Diamond penetrator* 150 kgf
77	77	—	30	—
79	79	—	34	—
83	83	—	38	—
87	87	—	42	—
91	91	—	46	—
95	95	—	50	—
100	100	—	54	—
106	106	—	58	—
112	111	39.0	62	—
119	118	41.0	66	—
126	125	43.0	70	—
135	134	45.5	74	—
145	144	47.5	78	—
157	155	50.0	82	—
171	168	52.5	86	—
188	184	55.0	90	—
209	204	57.5	94	—
234	228	60.5	98	20.0
248	241	61.5	100	22.5
285	275	64.5	—	28.5
326	313	67.5	—	34.0
362	346	69.5	—	38.0
404	382	71.5	—	42.0
452	425	73.5	—	46.0
481	450	74.5	—	48.0
513	—	75.5	—	50.0

Note: $\frac{1}{16}$ in ≡ 1.588 mm.

21.2 Tensile testing

21.2.1 Standard test pieces

Relevant standards: ASTM E 8-00 and E 8M-00, BS EN 10002-1 through -5.
The tensile test is the most commonly used mechanical test for metals. A specimen is extended under a uniaxial tensile force until separation occurs. The force required to deform the specimen is recorded as a function of extension, and from these data a number of intrinsic properties may be derived, which include: modulus of elasticity, yield and ultimate tensile strengths, ductility, and strain hardening behaviour. Further information regarding the tensile behaviour of metals and tensile testing may be found in Refs [6–9]. Tensile properties of common metals and alloys are presented in Chapter 22.

A wide variety of specimen geometries may be employed. For certain product shapes (e.g. strips and bars), a full cross-section of the product may be tested. In most cases, however, a part of the product is machined into a specimen which has a central section of reduced area and grip zones at either end. Cross-sections of the reduced area are usually circular or rectangular. In the reduced section, a gauge length is defined, from which measurements are made to determine ductility (Fig. 21.1).

In the European test standard, two types of 'proportional' test pieces are defined, regardless of the cross-section shape. For these specimens the gauge length L_0 and cross-sectional area A_0 are related by:

$$L_0 = 5.65\sqrt{A_0} \text{ and}$$
$$L_0 = 11.3\sqrt{A_0}$$

Non-proportional test pieces are also defined for various shapes.

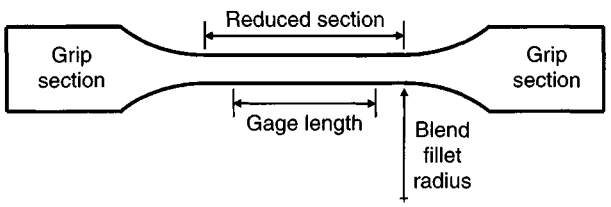


Figure 21.1 *Tensile test specimen*

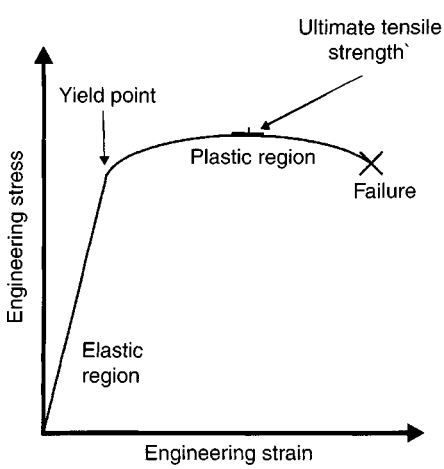


Figure 21.2 *Typical engineering stress–strain curve*

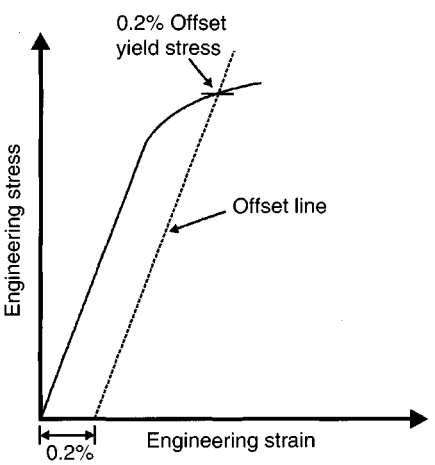


Figure 21.3 *Offset method for determining yield stress*

In the ASTM standard, specific dimensions are defined for the different cross-sectional shapes. The standard gage length and diameter for cylindrical specimens are 2.0 inch and 0.5 inch, respectively, while for rectangular cross-sections the gage length is 2.0 inch, the gage width 0.5 inch, and the thickness is variable. Sub-standard specimens proportionally based on the standard specimens are also defined. ASTM E 8M defines similarly sized specimens using metric units.

Several different grip end configurations are available. The specimen may be threaded, wedge-clamped, button-ended, or pin-loaded. The primary consideration for grips is that the specimen must not fail within the grip. The ability to maintain proper specimen alignment, so that the load is purely axial, is also important. Proper choice of grip arrangement is most critical for metals with limited ductility. If the specimen is machined from a product piece, the orientation of the long axis with respect to significant axes in the product (e.g. rolling direction) should be defined.

The force-extension data is gathered in a tensile test from transducers attached to the load frame and specimen. Strain-gauge based load cells built into the test frames are nearly exclusively used for force measurement. Extension is measured within the gauge length of the specimen, either mechanically (using clip-on strain gauges or linear variable differential transducers) or optically. Tensile behaviour is usually reported in terms of engineering stress and strain, which are calculated from the force and initial cross-sectional area (stress) and the extension and initial gauge length (strain). A typical stress–strain curve, shown in Fig. 21.2, shows an initial elastic region where stress and strain are linearly related, followed by non-linear plastic deformation at higher strains. Tensile behaviour may also be reported as true stress and strain, which are defined based on actual gauge length and area at a given applied force.

The strength of metal in a tensile test is quantified in terms of the yield strength and ultimate tensile strength (UTS). The yield strength reflects the stress needed to cause permanent plastic deformation; the exact value obtained from a given stress–strain curve depends on the definition used. The most common definition is the 0.2% offset strength, which is obtained from the intersection of the stress–strain curve and a line parallel to the initial elastic region but offset 0.2% in strain (Fig. 21.3). A second definition is the extension under load (EUL) which is defined as the stress needed to result in a given strain (typically 0.5%). The ultimate tensile strength is derived from the maximum load achieved in the tensile test, the highest point in the engineering stress–strain curve (see Fig. 21.2). For metals in which necking, a localised reduction in cross-section, occurs, the load on the specimen drops with continued extension beyond the UTS. The true stress in the neck does not diminish after the UTS is reached.

There are two measures of ductility assessed in a tensile test, elongation and reduction in area. Elongation is the increase in gauge length of the test piece after testing divided by the original gauge length. The increase may be manually measured by placing the two broken halves of the specimen back together and measuring the distance between gauge length marks placed on the specimen before testing, or by directly reading the extension at failure with an extensometer on the specimen. The two methods do not give equivalent results. Elongation values obtained are dependent on the initial gauge length of the specimen. This effect is particularly pronounced in materials which exhibit extensive necking. It is therefore important to report the gauge length along with tensile results.

Reduction in area is the difference between the original cross-sectional area and the smallest cross-sectional area after testing, expressed as a percentage of the original area. Reduction in area is not affected by gauge length differences, since measurements of area after testing are performed on any necked region.

The speed at which the test is conducted is an important parameter in tensile testing. Metal properties can exhibit a strong strain rate dependence, particularly at elevated temperatures. The strain rate is typically controlled through the rate of test machine cross-head displacement or the rate of stress increase, both of which can be roughly correlated with the strain rate. ASTM standard E 8 requires that the rate of stress increase be controlled to between 1.1 and 11 MPa/sec when determining the yield point. In any case, the speed of test (strain rate, stress rate, or cross-head displacement rate) should be reported with the tensile test data.

21.3 Impact testing of notched bars

Impact tests represent a simple way of measuring the toughness of a metal under rapid loading conditions and were the first fracture toughness tests devised. Despite the empirical nature of the data obtained, impact testing is still frequently used as a quality control measure and as a technique for determining the transition temperature from ductile to brittle behaviour in low-carbon steels. Further details on impact testing and the impact behaviour of metals may be found in Refs [6,10,11]. Data on impact energies for a number of common metals and alloys are presented in Chapter 22.

Table 21.8 DIMENSIONS OF IZOD IMPACT TEST PIECES—BRITISH STANDARD BS 131:PART 1:1961 (1996) AND AMERICAN STANDARD ASTM E23-01

	Square section		*Circular section*	
	Nominal dimension			
Item	mm	in *equivalent*	in	mm *equivalent*
Minimum overall length of test piece				
1 Notch	70[1]	2.75	2.8[1]	71
2 Notch	98	3.86	3.9	99
3 Notch	126[2]	4.96	5.0[3]	127
Width	10	0.394		
Thickness	10	0.394	0.45 dia.	11.48 dia.
Root radius of notch	0.25	0.010	0.010	0.25
Maximum depth below notch	8	0.315	0.32	8.1
Distance of plane of symmetry of notch from free end of test piece and from the adjacent notch	28	1.1	1.1	28
Angle of notch	45°		45°	

Notes: (1) American Standard overall length 75 mm (2.952 in).
(2) American Standard overall length 131 mm (5.157 in).
(3) American Standard overall length 137 mm (5.375 in).

21.3.1 Izod test

Relevant standards: ASTM E23-01, BS 131:Part 1:1961 (1996)
The test consists of measuring the energy absorbed in breaking a notched test piece by one blow from a striker carried by a pendulum. The test piece is gripped vertically with the root of the notch in the same plane as the upper face of the grips. The blow is struck on the same face as the notch and at a fixed height above it. Tests are usually performed at the ambient temperature of the test house.

TEST PIECES

The standard test pieces are either 10 mm square or 0.45 inches (11.4 mm) diameter cross-section. Complete dimensions are shown in Table 21.8. The notch profile is shown in Figure 21.2.

PRESENTATION OF RESULTS

When standard test pieces are used the following symbols are used in reporting the results of Izod tests:

I for Izod
S for square section
Rs for circular cross-section with straight notch
E.G. I120S: *x* ft lbf

An energy of *x* ft lbf was obtained from an Izod test with a striking energy of 120 ft lbf using a square section test piece.[†]

21.3.2 Charpy test

Relevant standards: ASTM E23-01, BS EN 10045-1:1990
The test consists of measuring the energy absorbed in breaking by one blow from a pendulum a test piece notched in the middle and supported at each end.

[†] BS 131:Part 2:1972 uses SI units with 1 J ≈ 4/3 ft lbf for striking energies (1 ft lbf = 1.355 82 J).

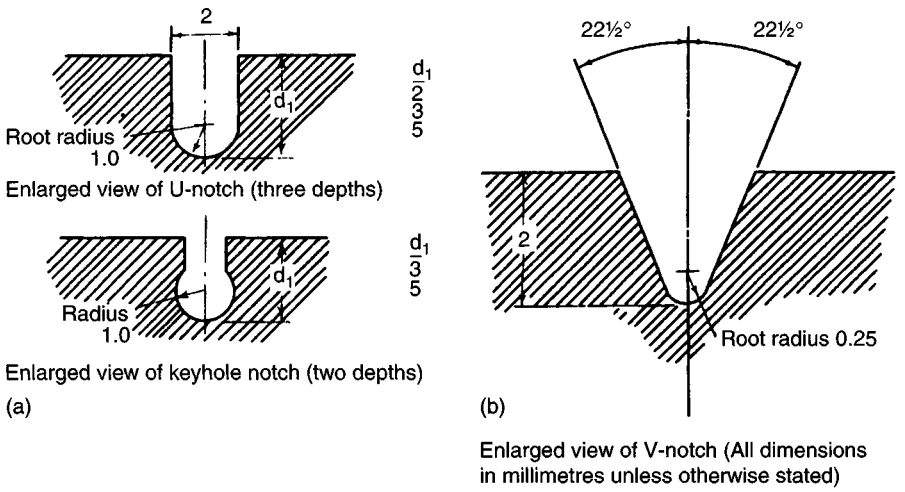


Figure 21.4 *Standard notch profiles for impact test pieces—see Tables 21.8 and 21.9*

Table 21.9 DIMENSIONS OF CHARPY IMPACT TEST PIECES—BRITISH STANDARDS BS EN 10045-1:1990; AMERICAN STANDARD ASTM E23-01

	Nominal dimension	
Item	mm	in *equivalent*
Length	55	2.165
Width standard test piece	10	0.394
subsidiary test piece[1]	7.5	0.295
subsidiary test piece	5.0	0.197
subsidiary test piece	2.5	0.098
Thickness	10	0.394
Root radius of V notch	0.25	0.010
Depth below V notch	8	0.315
Root radius of U notch	1	0.039
Depth below U notch	8 (2 mm notch)[2,3]	0.315
	7 (3 mm notch)[2,3]	0.276
	5 (5 mm notch)	0.197
Distance of notch from one end of test piece	27.5	1.083
Angle between plane of symmetry of notch and longitudinal axis of test piece	90°	
Angle of V notch	45°	

Notes: (1) Additional subsidiary test pieces are permitted by ASTM E23-01 as follows: 5 mm thick with 4 mm depth below the notch and 5, 10 or 20 mm width and 3 mm thick with 0.094 in (2.39 mm) depth below notch and 10 mm width.
(2) Not specified in ASTM E23-01.
(3) The 2, 3 and 5 mm notches are sometimes referred to as 'Mesnager', 'DVM' and 'Charpy', respectively.

TEST PIECES

The standard test piece has a 10 mm square cross-section with one of the three notch profiles shown in Figure 21.4. The test piece dimensions are shown in Table 21.9. Sub-standard test pieces are used where material thicknesses do not permit full-size specimens. It should be noted that the values obtained from subsidiary specimens cannot be compared with full-size specimens nor can the values obtained from different notches be compared. The current EN standard only includes the V- and U-notch profiles.

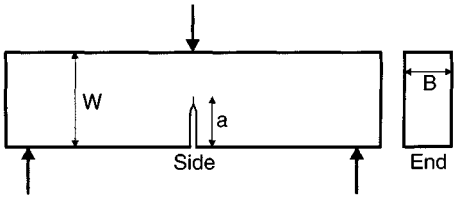


Figure 21.5 *Single-notch bend specimen*

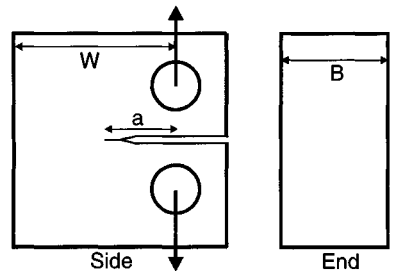


Figure 21.6 *Compact tension specimen*

PRESENTATION OF RESULTS

The report of the tests should include the following information:
Type of test: Charpy V or U notch; striking energy of the machine; size of test piece if sub-standard.
Nominal depth of notch and form ('U' or 'keyhole').
The energy absorbed (J) and the test temperature:

e.g. C160V : x J at y °C

An energy of x J was recorded from a Charpy V notch specimen tested at y °C with a striking energy of 160 J:

e.g. C320U2 : x J at y °C

In this case the specimen was a Charpy with a U notch 2 mm deep and the striking energy was 320 J.
Note: The above reporting form is used for the BS EN 10045-1 standard; there is no specified reporting form in ASTM E23-01.

21.4 Fracture toughness testing

Fracture toughness tests provide quantification of a material's resistance to crack extension using the principles of fracture mechanics in both linear-elastic and elastic–plastic forms. Tests are performed on specimens containing sharp, pre-existing defects formed by fatigue loading and are typically conducted at relatively slow loading rates (in contrast to impact tests, section 21.3). Information obtained in fracture toughness tests is used to determine the load-bearing capacity of materials and structures with defects. The following sections present information on common standardised fracture toughness tests for metals. Further information on fracture toughness testing is presented in Ref. [12]; the applicable standards must be conducted for complete test requirements.

21.4.1 Linear-elastic (K_{Ic})

Relevant standards: ASTM E 399-90, BS EN ISO 12737:1996
Linear elastic fracture toughness tests are used to measure resistance to crack extension for metals which are relatively strong and have relatively low toughness. For this case, the region of plastic deformation associated with the crack is small relative to the dimensions of the crack and the test specimen, and linear elastic fracture mechanics (LEFM) principles may be applied. In LEFM, the local stress field ahead of the tip of a sharp crack is characterised by the stress intensity parameter, K, which is dependent on the crack length, geometry, and applied load. The standard test measures K_{Ic}, the critical value of K for unstable crack propagation in an opening mode (mode I) with high constraint (plane-strain conditions). K_{Ic} is considered an intrinsic material property, and is believed to represent the lower limiting value of fracture toughness. For details on fracture mechanics and the applicability and use of K_{Ic}, consult Refs [13,14]. A limited amount of fracture toughness data is presented in Chapter 22.

Two specimen geometries are commonly used in fracture toughness testing, the single edge-notched bend and the compact tension specimens. These specimen names are abbreviated in ASTM standards as SE(B) and C(T), respectively; schematic diagrams of the two specimens are shown in Figs. 21.5 and 21.6. ASTM E 399 also defines three other specimen geometries with arc shapes

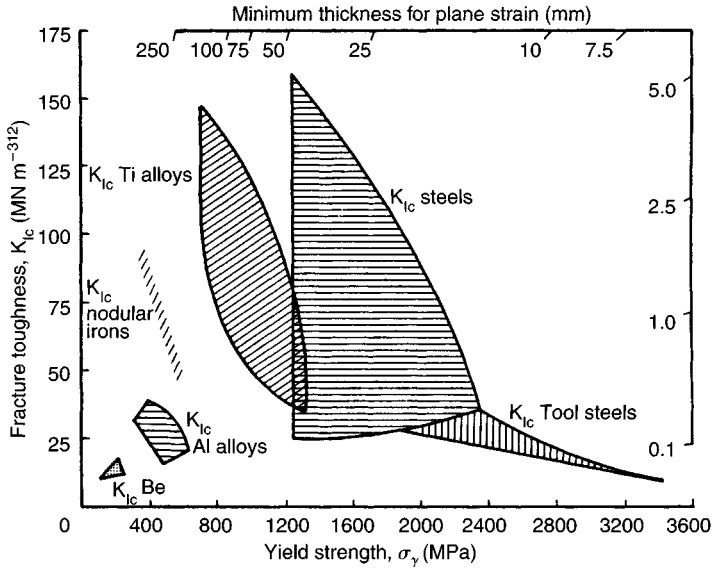


Figure 21.7 *Yield strength versus fracture toughness data for several alloy systems at 20°C. The upper scale of the diagram indicates the minimum thickness requirement for a given strength and toughness by projection of the line joining the origin and data point. (Source: Fulmer Materials Optimiser, Fulmer Laboratories Ltd, UK)*

Table 21.10 RECOMMENDED THICKNESS AND CRACK LENGTH FOR FRACTURE TOUGHNESS TEST PIECES

	Minimum recommended thickness and crack length	
σ_Y/E	mm	in
up to 0.005 0	100	4.0
0.005 0–0.005 7	75	3.0
0.005 7–0.006 2	63	2.5
0.006 2–0.006 5	50	2.0
0.006 5–0.007 1	38	1.5
0.007 1–0.008 0	25	1.0
0.008 0–0.009 5	13	0.5
0.009 5 or greater	6.5	0.25

that are less commonly used. All specimen types incorporate a sharp machined notch from which a fatigue crack may be grown. The SE(B) specimen requires more material for a given thickness than the C(T) specimen, but is simpler to machine.

The size of the specimen used is an important consideration, since certain size requirements must be met for the K_{Ic} test to be valid. The key dimensions of both specimen types are the thickness, B, and crack length, a (these are nominally equal). In order for a test to be valid, both must be larger than $2.5[(K_{Ic})/(\sigma_Y)]^2$, where σ_Y is the 0.2% yield stress for the material under the conditions of test, e.g. orientation, temperature, and loading rate. This requirement insures that the plastic zone ahead of the crack tip is sufficiently small with respect to these dimensions. Hence an initial measurement of σ_Y and an estimate of K_{Ic} must be made prior to selecting specimen size. Figure 21.7 can be used as a graphic guide to selection of thickness. In the absence of this information, the ratio of yield strength to elastic modulus may also be used to estimate specimen thickness (Table 21.10).

Prior to the actual fracture toughness test, a fatigue precrack must be grown from the starter notch. The standards define requirements for the length of fatigue crack which must be grown and the load

levels used. The objective of fatigue precracking is to obtain a microscopically sharp crack with an insignificant region of plastic deformation ahead of the crack tip.

The test itself involves monotonically increasing the specimen load until fracture occurs. The load and displacement are recorded and used to generate a plot of load versus displacement. Displacement is measured at the crack mouth using a clip gauge. After the test, the load-displacement record is analyzed to determine P_Q, a load from which K_Q is computed. K_Q serves as a provisional value of K_{Ic} until the validity requirements have been demonstrated. A secant line through the origin is drawn with a slope 5% less than the original linear region of the load-displacement curve. The load P_Q is defined as the intersection of the secant line with the load-displacement curve, or any higher recorded value prior to the intersection. K_Q is computed using the equation:

$$K_Q = P\, f(a/W)/B\sqrt{W}$$

Expressions for f(a/W) for the different specimen geometries are given in the standards. The crack length a is measured on the broken halves of the specimen at the end of the fatigue precrack.

A number of validity requirements must be met before the provisional K_Q may be accepted as K_{Ic}. In addition to the size requirements above (in which K_Q is substituted for K_{Ic}), the maximum load in the test must not exceed P_Q by more than 10%. This requirement limits the amount of stable crack extension. There are other requirements relating to the shape of the crack front and precracking load levels. The standards should be consulted for complete details.

21.4.2 K–R curve

Relevant standard: ASTM E 561-98

The K–R curve test provides a characterisation of fracture resistance for materials and specimen geometries in which stable, ductile crack extension occurs under elastic conditions. In the test, a plot of K, the stress intensity factor needed to drive the crack, as a function of crack extension, Δa, is developed, which is referred to as the R curve. An example R-curve is shown in Fig. 21.8. Once established, the R curve may be used to determine a value of K_c, the plane-stress fracture toughness, for various crack lengths and specimen geometries. The K–R test is frequently used to measure the fracture toughness of sheet materials, for which the thickness requirements (which ensure plane-strain conditions) of the K_{Ic} test (Section 21.4.1 above) may not be met. It is important to note that

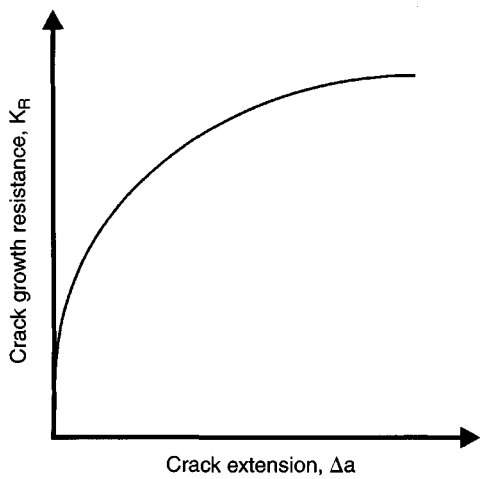


Figure 21.8 *Typical K–R curve*

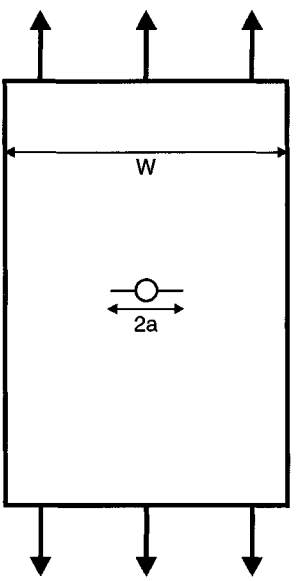


Figure 21.9 *Middle-cracked tension specimen*

results obtained in K–R curve testing are dependent on specimen thickness—therefore all results are given for a specified thickness.

The test involves loading a notched specimen with a fatigue precrack in tension to cause stable crack growth. The load is plotted as a function of crack mouth displacement. The effective crack length (accounting for plasticity ahead of the crack tip) is determined either by physical measurement or from the load-displacement record. The load and effective crack length are used to calculate K for a series of increasing effective crack lengths, which are then plotted as a function of crack extension (total effective crack length minus initial crack length). Specimens with the compact tension C(T) or middle cracked tension M(T) geometries may be used with this technique. The M(T) geometry is used for sheet materials—a schematic diagram is shown in Fig. 21.9.

An alternative technique for K–R curve determination allowed in the standard is the use of a crack-line-wedge-loaded compact specimen C(W). The geometry is similar to the C(T) specimen, except that the loading is applied via a wedge driven into the crack mouth, resulting in displacement-controlled loading. This loading technique is useful for lower-toughness materials. Full details on the use of this specimen may be found in the standard.

Data obtained in R-curve testing under this standard must be principally in the elastic loading regime. This requirement is expressed for the C(T) and C(W) specimens as:

$$b = (W - a) \geq (4/\pi)/(K_{max}\sigma Y)^2$$

where b is the uncracked ligament and K_{max} is the maximum level of K reached in the test. For the M(T) specimen the net section stress based on the uncracked ligament must be less than the yield stress.

21.4.3 Elastic-Plastic (J_{Ic}, CTOD)

Relevant standards: ASTM E 1820-01, BS 7448-1:1991, BS 7448-4:1997

For low strength, high-toughness materials (such as low-carbon steels) the requirements of LEFM (K_{Ic}) testing are not easily satisfied as extensive ductile crack extension with associated plasticity occurs prior to failure. To quantify the toughness behaviour in this regime, elastic-plastic fracture mechanics tests have been developed. Several separate tests which use similar specimen geometries and techniques exist. These tests have been recently consolidated by the ASTM (along with K_{Ic} testing) into a single test standard, E 1820-01. Two British Standards covering the same test procedures exist. The tests provide different measures of the conditions for initiation of stable crack extension (similar to K_{Ic}) and the resistance to further crack extension (R-curve behaviour). Details of the tests are found in the standards and Ref. [12].

The two parameters used in elastic-plastic fracture toughness testing are the J integral and the crack tip opening displacement (CTOD or δ). The J integral is a non-linear measure of the driving force available to extend a crack,[14] and is analogous to G, the strain energy release rate in the linear regime. The J integral is useful because it can be accurately calculated from experimental data or using numerical methods. The CTOD is the opening at the position of the original crack tip, a parameter that is believed to characterise the fracture process.[13] The CTOD is calculated from the measured displacement at the crack mouth (CMOD), which is simple to measure experimentally. Due to simplicity, the CTOD was the first parameter used to measure the fracture behaviour of materials beyond the LEFM regime. Its use, however, has to a large extent been superseded by the J integral, because of the assumptions which must be made in calculation of CTOD from CMOD.

Two specimen geometries are allowed in the standards—the C(T) and SE(B) specimens, both essentially identical to the geometries used for K_{Ic} testing. The notch of the C(T) specimen is widened for J-integral testing to allow for displacement measurements to be made along the load line. In addition, specimens used in J-integral testing may incorporate side grooves to keep the crack in its original plane. All specimens are tested after creation of a sharp precrack using fatigue loading, as is performed in K_{Ic} testing.

The basic test procedure for both J-integral and CTOD tests involves loading the precracked specimen while monitoring the CMOD and/or load-line displacement until unstable fracture occurs or a desired displacement level is reached. Depending on the behaviour of the material and the information desired, a number of different fracture parameters may be obtained. The exact test and analysis procedures are complex, and the applicable standards must be consulted for details.

The fracture toughness parameters measured in J-integral or CTOD testing are useful for comparing the behaviour of different materials or as a quality control measure, and may also serve as a basis for defect tolerance assessment of structural materials. However, particular caution must be used when applying data obtained with laboratory test specimens to actual service conditions.

21.5 Fatigue testing

Fatigue tests measure the resistance of a material to failure by application of cyclic stresses. A very broad range of stresses and number of cycles to failure are encountered, from less than 100 cycles for stresses well in excess of the yield strength, to millions of cycles for low, elastic stresses. Historical tests have used specimens with smooth surfaces or relatively blunt notches in which the number of cycles to initiate and propagate a crack resulting in failure were measured. With the advent of fracture mechanics and damage-tolerant design principles, tests to measure sub-critical crack propagation rates under cyclic loading were developed and are commonly used today.

This section presents basic information concerning fatigue tests which have been standardised, describing properties measured, specimen and machine requirements, basic test procedures, and important considerations in testing. The reader should always refer to the relevant test standard for complete details. For further information concerning the fatigue behaviour of metals, consults Refs [15–17]. Fatigue data for a number of common metals and alloys are presented in Chapter 22.

21.5.1 Load-controlled smooth specimen tests

Relevant standards: ASTM E 466-96, BS 3518-12:1962 (1997), BS 3518-3:1963 (1997)

Load-controlled fatigue tests are typically performed at stresses in the elastic range, resulting in relatively long fatigue lives (greater than 10^5 cycles to failure). The specimen surfaces are polished smooth or contain machined notches. Each fatigue test measures the number of cycles to failure at a single stress level; failure is nearly always defined as complete specimen separation. Several fatigue tests at different stress levels are required to define the fatigue behaviour in terms of a S-N curve, a plot of cycles to failure (x-axis) versus stress range (y-axis). The curve defines stress levels below which failure in a given number of cycles will not occur. The S-N curve may be used to derive a 'fatigue strength', or stress level below which failure does not occur in a very high number of cycles, usually 10^7 or 10^8.

Load-controlled fatigue tests are typically performed at cyclic frequencies greater than 10 Hz and sometimes as high as 1000 Hz to achieve failure in reasonable times. Loading may be either in simple tension or rotating bending. A number of different machine designs are commercially available, including servo-hydraulic and electromechanical. Rotating-bend tests are favoured for high frequencies, since these are easily achieved and the test machine is relatively simple and inexpensive, allowing a number of machines to be obtained and used simultaneously. Very high frequency tests ($\sim 10^4$ Hz) can be performed using ultrasonic excitation of the specimen.[18]

The applied load usually follows a sinusoidal waveform. The maximum and minimum loads in the cycle define the mean load applied, and the load ratio, R ($=P_{max}/P_{min}$). In the case of fully-reversed (equal tensile and compressive, $R = -1$) loads the mean load is zero. Most S-N curves are produced with fully-reversed loading; other types of diagrams may be constructed to account for the effect of non-zero mean loads.[17] Note that the waveform for rotating-bend tests is always sinusoidal and fully-reversed.

Dogbone-shaped specimens very similar in shape to the tensile specimens described in section 21.3 are defined in the standards. Cross-sections may be circular (necessary for rotating bend) or rectangular. The gauge sections of fatigue specimens need not be straight, but may incorporate a shallow radius as shown in Fig. 21.10. Because fatigue cracks are frequently surface-initiated, the

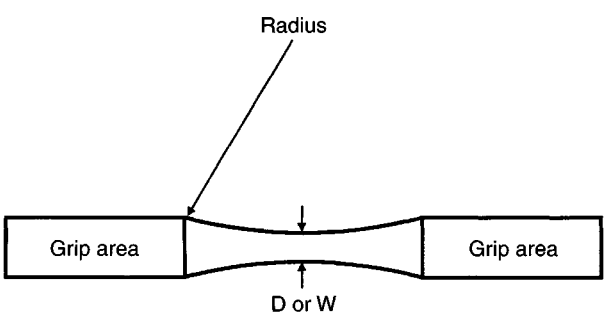


Figure 21.10 *Strain-controlled fatigue test specimen with continuous radius between ends*

finish of the specimen is very important. The standards define allowable machining removal rates and final polished roughness. The grip arrangement for fatigue specimens is also important, more so than in tensile tests. Threaded grip attachments are not recommended since the notches may initiate cracks.

Specimen alignment is critical in axial fatigue tests. The load must purely axial without the presence of significant bending stresses. Limits on the amount of bending stress are defined in the standards. Alignment is usually verified by loading a dummy specimen with several strain gages on the surface and comparing strain readings.

Load-controlled fatigue tests may be conducted at any desired temperature, provided that the gauge section of the specimen is uniformly heated or cooled. Several different furnace designs are available for use with fatigue testing machines, including clamshell resistance furnaces, induction furnaces, and radiative (lamp) furnaces. If the grips are heated or cooled along with the specimen they must be constructed from suitable materials. Temperature measurement of the specimen must be performed with care, so that surface defects are not introduced by the attachment of thermocouples.

The actual test procedure is simple, involving placing the specimen in the grips, applying a known cyclic force, and recording the elapsed cycles until failure occurs. A number of variables may influence the results obtained in load-controlled fatigue testing, and it is important to ensure that the conditions of test closely match those of the intended application. Variables to consider include: temperature, environment, humidity, specimen size, frequency, surface finish, and stress concentrators. Because of the statistical nature of fatigue phenomena, several specimens should be tested for each condition to assess the extent of scatter. This is especially important for lower-stress, longer-life conditions.

21.5.2 Strain-controlled smooth specimen tests

Relevant standards: ASTM E 606-92, BS 7270:1990

Tests in the so-called low cycle fatigue (LCF) regime usually involve significant plastic deformation of the specimen and are conducted with axial loading in strain control using servohydraulic testing machines. The use of strain as the test parameter is accomplished using extensometers attached to the specimen for feedback control of the test machine. Details on extensometry and testing machines used for LCF testing may be found in Refs [19–21]. The tests are otherwise similar to load-controlled tests, in that each specimen gives a single data point of cycles to failure at the applied strain range. Failure is not always defined as complete specimen separation, however, but may also be a specified drop in maximum load, observation of crack initiation, or a change in modulus ratio. Plots of cycles to failure as a function of applied strain range are constructed.

In addition to the number of cycles to failure, the cyclic stress–strain behaviour of the specimen may also be reported. This takes the form of hysteresis loops, plots of stress versus strain for a single fatigue cycle (Fig. 21.11). Changes in the hysteresis loop from the beginning to end of the test indicates whether material hardens or softens under cyclic loading. Instead of full hysteresis loops, the maximum and minimum loads may also be recorded. The load range at 50% of the cycles to failure is commonly reported.

LCF tests are typically performed at relatively low frequencies, less than 1 Hz, and at elevated temperatures. Strain waveforms are usually triangular rather than sinusoidal, so that loading is accomplished at a constant strain rate. Strain-rate effects are important at elevated temperatures due to the addition of creep as a deformation mechanism. Straining in LCF tests is nearly always fully-reversed, with equal tensile and compressive strains. The loads needed to impose these strains may not be equal. Smooth specimen LCF tests may be used to measure creep-fatigue behaviour, in which hold periods of constant strain are imposed in which creep occurs. Details of creep-fatigue testing may be found in Refs [20,21].

Specimens used in LCF testing must be designed in conjunction with the test machine gripping arrangements to ensure that the load profile in the transition from tension to compression is continuous. The grip ends must be securely clamped. As with load-controlled tests, surface finish is important, although it may be less so if significant specimen oxidation occurs at elevated temperatures during testing. Alignment of the specimen is again critical.

21.5.3 Fatigue crack growth testing

Relevant standards: ASTM E 647-00, BS 6835-1:1998

Fatigue crack growth testing measures rate of advance of a fatigue crack in terms of the applied driving force for growth using Linear Elastic Fracture Mechanics (LEFM) principles.[13–15] The range

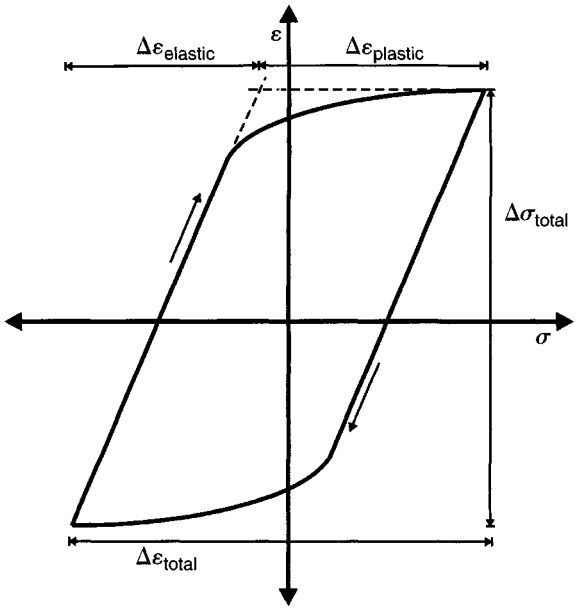


Figure 21.11 *Typical hysteresis loop obtained in strain-controlled low-cycle fatigue testing*

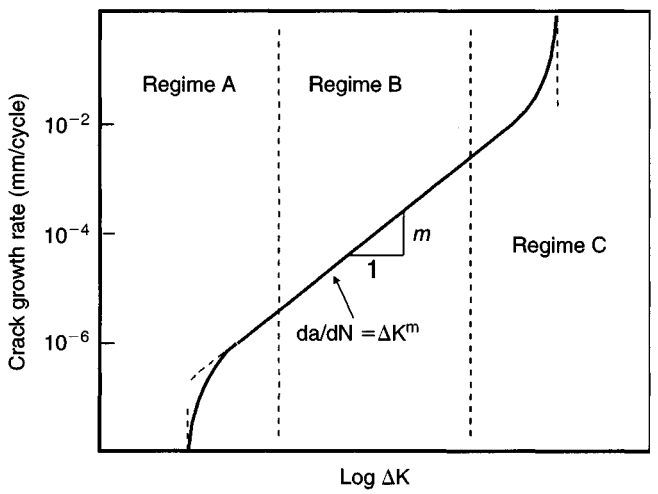


Figure 21.12 *General plot of crack growth rate versus stress intensity range showing three regimes of crack growth*

of stress intensity factor in the loading cycle (ΔK), is used as the driving force parameter. As described in Section 21.4, K is proportional to the applied load, the square root of the crack length, and a geometric factor. ΔK is simply the difference between the maximum and minimum K in each load cycle, and hence is proportional to the applied load range. The rate of crack growth, da/dN, where a is the crack length and N is the cycle number, is measured over a range of ΔK and a plot constructed. A generic plot of da/dN versus ΔK typical of metals is shown in Fig. 21.12. There are three characteristic regimes, the near-threshold region at low ΔK, the Paris regime where the logarithm of da/dN is linearly dependent on ΔK, and the stage III regime of high crack growth

rates near failure. A threshold value of ΔK below which crack growth does not occur is commonly observed for metals. Further information on the fatigue crack growth behaviour of metals can be found in Ref. [15].

The fatigue test standards describe specimen, test machine, and reporting requirements for measurement of fatigue crack growth rates above 10^{-8} m/cycle. In the ASTM standard, procedures for measurement of fatigue threshold values are also given.

The specimens used in fatigue crack growth testing are identical to those used in other fracture mechanics tests (see section 21.4). In the ASTM standard three specimen types are defined: the compact tension, the single-edge bend, and the center-cracked tension. Similar specimen geometries are defined in the British standard. The specimens contain machined, sharp notches from which fatigue cracks are grown. Prior to the acquisition of valid crack growth rates, the specimens must be precracked to beyond the region of disturbed material at the notch root. Limits are set on the ΔK which may be used for precracking. Standard servo-hydraulic testing machines are used for crack growth testing, frequently equipped with computerised control and data acquisition systems to enable automated testing.

A critical aspect of fatigue crack growth testing is precise measurement of crack length. Three techniques are commonly used for crack length measurement: optical, compliance, and potential drop. In the optical technique the crack is measured at intervals using a travelling microscope. Strobe lighting may be used to allow easy observation of the crack while cycling. This is a historical technique not frequently used today due to the labour involved and lack of very fine resolution needed for threshold measurements. It is simple, however, and does not require computerised equipment.

The optical technique has been largely supplanted by compliance and potential drop techniques. In the compliance technique, the compliance of the specimen is computed from the crack-mouth opening displacement (measured with a clip-gauge mounted to the crack mouth) and the applied load. A calibrated curve of compliance as a function of crack length for the particular specimen geometry is used to derive crack length. In the potential drop method, a large DC current is passed through the specimen, and the potential generated across the crack mouth is measured. The potential drop increases with crack length, which can be quantified, again through a calibration curve. These techniques have the advantage of providing an accurate, essentially continuous measurement of crack length and are highly suited to automated control of the fatigue test. In fact, the complex nature of crack length derivations make automated, computerised measurement essentially a necessity.

Fatigue crack growth data in the higher growth rate regime (greater than 10^{-8} m/cycle) may be simply acquired by growing the crack at a constant load range expected to result in an initial growth rate near 10^{-8} m/cycle. ΔK increases as the crack grows, allowing a wide range of data to be obtained. Fatigue crack growth rates are calculated from the applied load range and crack length versus cycle number data using incremental polynomial or secant techniques.

To obtain near-threshold crack growth rate data (below 10^{-8} m/cycle), ΔK is decreased as the crack grows, a procedure referred to as 'load-shedding.' Proper load-shedding is complex because of three factors: ΔK depends on the crack length; the crack must be grown through the plastic zone of the previous ΔK level before the next reduction; and the threshold level must be reached before the crack has grown through the specimen. These considerations are accounted for in ASTM E 647-00; computerised, automated testing capability is again essential.

The frequencies used in fatigue crack growth testing are typically greater than 10 Hz to allow testing in reasonable timeframes. Fatigue crack growth rates may be a function of several material, environment, and test variables, depending on the metal and the regime of crack growth measured. A few examples include grain size, orientation, humidity, and load range. Requirements for analysing crack growth data and full reporting of fatigue results are given in the relevant standards. Fatigue crack growth testing may be performed at elevated temperatures or specific environments. Compliance and potential drop crack measurement techniques are especially amenable to testing in these situations.

21.6 Creep testing

Relevant standards: ASTM E 139-00, BS EN 10291:2000

Creep is defined as the time-dependent elongation of a metal which occurs at static stresses less than the nominal yield strength. For most metals creep only occurs at temperatures exceeding half of the absolute melting point. Creep testing measures the rate of specimen elongation as a function of applied stress, temperature, and test time. A number of tests must be conducted to fully characterise the behaviour of a given material. Details on the creep behaviour of metals may be found in Refs [22–24], and creep properties of common metals and alloys are presented in Chapter 22.

There are three types of creep tests: creep, creep-rupture, and stress-rupture tests. In a creep test, testing is only performed as long as necessary to obtain the minimum creep rate at the specified stress and temperature; the test is not performed to failure. Creep-rupture refers to a creep test conducted to failure. Creep rates are not measured in stress-rupture testing, only the time to failure under creep loading conditions. All three tests use similar specimens and load frames.

The specimens used for creep testing are essentially identical to those used in simple tensile tests (see Fig. 21.1), although creep specimens may have grooved attachment points on the shoulders of the specimen and longer gauge lengths to permit more accurate strain measurement. Creep testing may also be performed in compression; in this case the specimens may be either cylindrical or parallelepipeds with aspect ratios between 2 and 4.

Creep test frames are simple in design, with a static load applied with weights through a typically 20:1 lever arm. Clamshell-type resistance furnaces are used for heating. Due to the common use of high temperatures (frequently greater than 800°C), extensometers for creep testing are designed to translate the specimen extension out of the hot zone, where measurement is made using standard LVDT transducers. Pull rods, gripping devices, and extensometers must be constructed from suitable high-temperature materials. Due to the long test durations and the necessity to perform several tests to cover a range of stresses and temperatures, a creep testing laboratory usually includes a large number of frames, often with a central data collection system.

The test standards define specimen, load verification, and data reporting requirements for all types of creep tests. As with other axially-loaded mechanical tests, specimen alignment is critical and must be verified to demonstrate that bending stresses are negligible. An important additional parameter to consider in creep testing is the environment, which may have a strong influence at the high test temperatures commonly encountered. Specially designed retort systems are commercially available to enable testing in either inert or vacuum environments.

21.7 Non-destructive testing and evaluation

Non-destructive testing and evaluation (NDT, NDE) techniques permit examination of metal parts or samples without alteration or destruction. NDE encompasses all phases of the production, use, and eventual rejection of components and structures. The initial scope of NDT was the detection of damage in the form of flaws, cracks, inclusions, etc. Since the primary concern has been detection of the smallest possible flaw, sensitivity and probability of detection are important aspects of NDT technology. For materials which have many flaws, however, detecting an individual flaw is immaterial and the overall effect of the flaws on the residual strength of the structure is of greater importance. NDE has been more recently used for process control in manufacturing; a complex application since the inspection equipment must be compatible with process conditions, production equipment, and environment.

Another aspect of NDE, generally not considered, is the integration of NDE into the engineering design process.[25] Despite advances in NDE, it is still primarily used as an after-the-fact testing tool. There are numerous examples where neglecting NDE during the design stage has made its eventual use more costly. Components to be examined are either difficult to access or the available NDE tools are inadequate to perform an exhaustive evaluation. NDE must be an integral part of design to overcome these drawbacks. Various NDT & E techniques and their detection capabilities are listed in Table 21.11.

Table 21.11 BASIC NDT & E TECHNIQUES AND THEIR DETECTION CAPABILITIES

ND technique	Detection capability
Ultrasonic	Cracks, voids, inclusions, corrosion, anisotropy, thickness, elastic constants, grain size, porosity, crack propagation
Radiography	Cracks, voids, density variations, inclusions
Electromagnetic	Cracks, voids, alloy contents, inclusions, corrosion, anisotropy, electrical and thermal conductivity, thickness, heat treatment
Acoustic emission	Cracks, inclusions, plastic zone formation, damage propagation monitoring, leak detection
Thermal	Cracks, inclusions, voids, grain size, porosity, thermal conductivity, composition
Microwave	Thickness measurements, cracks, distributed inhomogeneity

21.7.1 Ultrasonic

Ultrasound is defined as acoustic waves with frequencies above the range of human hearing (>18 MHz). Wave velocity and attenuation are the two parameters measured in ultrasonic examination. Material discontinuities (e.g. voids, cracks, or inclusions) separate an impinging ultrasonic wave into three components: transmitted, reflected, and scattered. Each of these signals carries information about the material, such as stiffness, density, flaw location, and size[26]. An accurate measurement of the time of flight of the wave provides the exact location of the flaw. The reflected intensity is proportionate to the size of the flaw, therefore, measurement of the reflected signal amplitude and its spread effectively estimate the size of the flaw. Basic ultrasonic configurations are shown in Fig. 21.13.

The following relate the wave velocity of the sound to the stiffness E and Poisson's ratio ν of the material:

$$E = \frac{\rho c_L^2(3c_L^2 - 4c_s^2)}{c_L^2 - c_s^2} \quad \text{and} \quad \nu = \frac{c_L^2 - 2c_s^2}{2(c_L^2 - c_s^2)}$$

where ρ is the density of the medium and c_L and c_S are the longitudinal and shear wave velocities in the medium, respectively. E and ν can be calculated if the two wave velocities are measured accurately.[27,28] Since time, distance, and velocity are related, in a simple experiment at least one of the variables must be known. A variation of this technique, which simultaneously gives both velocity and thickness, is reported in Refs. [29,30]. Ultrasound can also be used to investigate anisotropic materials where the velocity measurements depend on the direction of the wave propagation.[31,32] Waves travelling in the plane of a plate, called Lamb waves or plate waves, are used to measure the elastic properties in the plane of the plate.[33] The scattered acoustic signal has been used to characterise microstructural features in titanium alloys.[34]

21.7.2 Radiography

Radiography is one of the most widely used NDE techniques.[35] The transmission of radiation (x-rays, gamma-rays, or neutrons) through a material is recorded on a photographic plate. Variation in the transmission through the material (due to inhomogeneity, defects, or thickness variations) results in contrast on the film. In determining defect size, magnification in the radiograph due to perspective must be accounted for, as shown in Fig. 21.14.

The radiograph image quality is defined by three factors: contrast, clarity, and graininess. *Contrast* results from varying transmission of the radiation through the object. Higher contrast images are easier to interpret. The edge *clarity* of the imaged feature depends on its size distance from the radiation source. The edges of the feature produce a shadow relative to the finite size of the source resulting in blurred boundaries, as shown in Fig. 21.14. In the last 15 years, advances in radiographic equipment and digital data processing have enabled the detection of features to reach new thresholds. The micro-focus x-ray source with a spot size of 5 microns has resulted in a significant improvement in resolution and clarity. The *graininess* of an image is a property of the imaging film. These fine grain-like spots can decrease the ability to detect small flaws. Digital recording and processing of data and digital enhancement can overcome these problems.[36]

An x-ray simulator (XRSIM) has been recently developed that simulates the entire process of the radiography, starting with a computer model of the object. This simulation aids the operator in deciding proper inspection angles and equipment calibration. The simulation can make NDE an

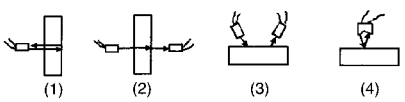


Figure 21.13 *Basic ultrasonic configurations: (1) reflection, (2) transmission, (3) surface or plate, (4) acoustic microscopy*

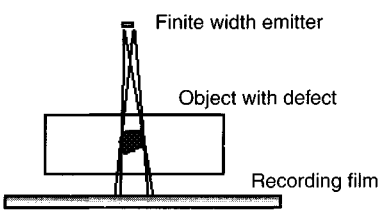


Figure 21.14 *Image magnification and shadow effect in radiography*

integral part of component design process, allowing NDE technicians to work with model developers to ascertain that components are easy to test during service.

21.7.3 Electrical and magnetic methods

The fundamental electrical and magnetic properties of materials are electrical conductivity, electrical permittivity, and magnetic permeability. These properties can be related to other characteristics such as microstructure, strength, hardness, and the presence of flaws and inclusions.[37,38] The primary electrical and magnetic NDE testing methods are: magnetic particle, magnetic flux, and eddy current testing.

MAGNETIC PARTICLE TESTING (MT)

This method is accomplished by using a yoke or a coil to induce a magnetic field, or by passing a current through a ferromagnetic material, and then dusting the surface with iron particles (either dry or suspended in liquid). Surface imperfections (defects such as cracks) distort the field. Concentration of iron particles due to field distortion reveals the defects.

MAGNETIC FLUX METHODS

Magnetic field distortion due to defects can be quantitatively assessed with devices such as Hall probes, magnetoresistive sensors, or coils.

EDDY CURRENT METHOD

The principal advantage of the eddy current method is that it does not require surface preparation, as there is no contact between the probe and the object under study.[39] Examples of feature which may be detected and measured using the eddy current method include surface and sub-surface cracks, the thickness of metallic plates, paint thickness on metal surfaces, and corrosion extent. Eddy currents are generated by an alternating magnetic field induced in the specimen with alternating currents in a coil separate from the specimen. The orientation of the coil is determined based on the assumed defect plane to obtain maximum sensitivity of detection. Imperfections or changes in the specimen's conductive properties will interrupt the flow of eddy currents, causing changes in the induced magnetic field that can be recorded.

21.7.4 Acoustic emission testing (AE)

The use of AE testing in the study of metals encompasses detection of crack propagation, yielding and plastic deformation, corrosion, and creep. AE is also used for continuous monitoring of structures like bridges, pipelines, dams, and cables, and for detection of in-service damage or leaks. When a metal is stressed, imperfections or cracking events emit short bursts of acoustic energy–high-frequency (100–300 kHz) transient elastic waves, which may be detected using piezoelectric sensors attached to the specimen. The acoustic emission technique is very sensitive and can easily detect defects that cannot be optically resolved.[40,41] The frequency and amplitude of the acoustic waves can be used to differentiate between different types of events and defects. For example, waves emitted during plastic deformation have low frequencies compared to those emitted during cracking.[42]

21.7.5 Thermal wave imaging

This method is useful in detecting subsurface variations of thermal properties, defects such as cracks or inclusions, and spatial variation of material properties.[43–45] It involves a burst of energy in the form of a flash lamp or a laser light that is impinged on the object under study. The resulting thermal waves are recorded using an infrared lens.

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